

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 6: Rotating-Frame Corrections

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Overview

The rotating-frame sector introduces the minimal quadratic corrections needed to reproduce disk and clock comparisons.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$g_{00}^{rot} = g_{00} + \mu_1 \Omega^2 r^2 / c^2$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$g_{\varphi\varphi}^{read} = g_{\varphi\varphi}(1 + \mu_2 \Omega^2 r^2 / c^2)$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$\mu_1 = 1, \mu_2 = -1/2$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper C, §§1–3
- Rotation Registry R-01–R-08
- SGT9 patch matrix P-2